PREDICTIVE MAINTENANCE OF VEHICLES USING MACHINE LEARNING

METHODOLOGY:

The proposed methodology aims to develop a reliable machine learning-based predictive maintenance framework for logistics fleet vehicles by utilizing historical maintenance records, operational parameters, environmental conditions, and telematics information. The methodology follows a systematic sequence of data preparation, feature analysis, model development, and performance evaluation to ensure accurate and reliable maintenance prediction.

The study begins with the collection of the **Logistics Vehicle Maintenance History Dataset** obtained from Kaggle. The dataset consists of approximately **250,000 vehicle records** with **50 attributes**, including vehicle specifications, maintenance history, operational data, weather conditions, road conditions, engine health parameters, telematics information, and diagnostic indicators. The target variable, **Maintenance_Required**, represents whether a vehicle requires maintenance and forms the basis for the prediction model.

Initially, the dataset is examined to understand its overall structure and quality. Data types, missing values, duplicate records, descriptive statistics, and class distribution are analyzed to identify potential issues before model development. This step ensures that the dataset is suitable for machine learning and provides a clear understanding of the available information.

Following dataset understanding, **Exploratory Data Analysis (EDA)** is performed to investigate the distribution of numerical and categorical features, identify relationships between variables, detect outliers, and analyze correlations with the target variable. The insights obtained from EDA assist in selecting relevant features and understanding the characteristics of maintenance-related data.

A **feature leakage analysis** is then conducted to identify variables that may directly or indirectly reveal the target variable during model training. Features derived from maintenance outcomes or future information can produce artificially high prediction accuracy and reduce the model's applicability in real-world scenarios. Therefore, leakage-prone attributes are identified and removed before further processing.

Once leakage analysis is completed, **feature selection** is performed to eliminate redundant and irrelevant variables while retaining the most informative predictors. The selected features are then preprocessed by encoding categorical variables and preparing numerical attributes for machine learning algorithms. Where necessary, feature scaling is applied to improve model performance.

The prepared dataset is divided into **training (80%) and testing (20%)** subsets. To study the effect of class imbalance, two independent experiments are designed. The first experiment utilizes the original dataset without modification. The second experiment applies the **Synthetic Minority Oversampling Technique (SMOTE)** to the training data only, thereby balancing the minority class while preventing information leakage from the test set. This comparative approach enables a fair assessment of the impact of class balancing on predictive performance.

Several supervised machine learning algorithms, namely **Logistic Regression, Decision Tree, Random Forest, and XGBoost**, are trained using both experimental datasets. These algorithms are selected because they are widely used for structured classification problems and provide a balance between interpretability and predictive performance.

The developed models are evaluated using multiple performance metrics, including **Accuracy, Precision, Recall, F1-Score, ROC-AUC Score, and Confusion Matrix**. Using multiple evaluation metrics ensures a comprehensive assessment of model performance, particularly in the presence of class imbalance where accuracy alone may not provide reliable conclusions.

After evaluation, the best-performing model is further optimized through **hyperparameter tuning** to improve prediction accuracy and generalization capability. The optimized model is then serialized using the **Joblib** library, enabling future integration into a web-based predictive maintenance application for logistics fleet management.

The proposed methodology provides a systematic and reliable framework for developing an intelligent predictive maintenance system capable of assisting logistics organizations in reducing maintenance costs, minimizing unexpected vehicle failures, improving fleet availability, and supporting data-driven maintenance decision-making.